INCH-POUND

MIL-W-44027A
3.2 May 1993
SUPERSEDING
MIL-W-44027
18 June 1980

MILITARY SPECIFICATION

WEDGE KIT,RAMP UNDLE DELIVERY SYSTEM, C-130 AIRCRAFT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for wedge kit, of one type and size for a ramp bundle delivery system (see 6.1).

2, APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standard, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of specifications and Standards (DODISS) and Supplement thereto, cited in the solicitation (see 6.2).

(Beneficial comments recommendations, additions, deletions) and any pertinent data may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5017 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 1670

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributed
MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-2073-1	-	DoD Materiel Procedures for Development and Application of Packaging Requirements
MIL-STD-2219	-	Fusion Welding for Aerospace Applications

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardiaztion Documents Order Desk, Building 4D, 700 Robbbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

32-1-2639	Wedge Kit, Ramp Bundle Delivery System, C-130 Aircraft

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-UXT, Natick, MA 01760-5017.)

2.2 Non–Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documentations which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the document cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 165	- Liquid Penetrant Inspection Method

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organization that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and references cited herein, the text of this document takes precedence. Nothirg in this document, however, supersedes applicable laws & regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjectsd to first article inspection (see 6.3) in accordance with 4.3.

3.2 Components and materials. Components and materials shall conform to the requirements on the applicable drawings, to subsidiary specifications and standards applicable thereto, and as specified therin. It is encouraged that recycled material be used when practical as long as it meets the requirments of this specification.

3.3 Construction. Construction details shall conform to the requirements of Drawing 11-1-2639 and to the subsidiary specifications and standards applicable thereto.

3.3.1 Welding. welding of the aluminum components shall conform to the requirements of MIL-STD-2219.

3.3.2 Painting. Preparation and painting shall be as specified on the individual drawings.

3.3.3 Marking. Marking shall be as indicatsd on the individual drawings and Shall conform to MIL-STD-130.

3.4 Workmanship. Workmanship shall conform to the quality established by this document and the applicable drawings. Cut edges shall be smooth, free of burrs, sharp edges and corners, and shall be neatly finished. Bolts and nuts shall be drawn tight and so secured that they will not work loose under vibration.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise speified in the contract or purchase order, the contractor is responsible for the performance of all inspeciton requirements (examinations and tests) as specified herein. Except as otherwise specified in the oontract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements speciied herein, unless disapproved by the Government. The Goverment reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an asseptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.1 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classificaiton of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4. As part of the first article, unfinished weld samples, representing each type of weld, shall be inspected in accordance with 4.4.2.2. The presence of any defect or failure to pass any test shall because for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the in accordance with all the requirements of referenced documents unless wtherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.2 Certification. The contractor shall furnish a certificate of compliance ceftifying that the paint used is in accordance with the requirements of the applicable drawings.

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4.4.2 In-process inspection. Inspection of subassemblies shall be made to ascertain that construction details which cannot be examined in the finished product are in accordance with specified requirements. The Government reserves the right to exclude from consideration for acceptance, any material or service for which in-process inspection has indicated noncomformance.

4.4.2.1 Visual examination of operations. Examination shall be made of the following operations to establish conformance to specified requirements.

a. Verigy that welding is performed in accordande with MIL-STD-2219 (see 3.3.1).

b. Verify that surfaces to be painted have been properly prepared for paintig and that prim and finish coat are applied in accordance with prescribed specifications (see 3.3.2).

c. Verify that plating is performed in accordance with the specification cited on the drawing.

4.4.2.2 Liquid penetrant weld inspeciton. Prior to any surfoce treatment, all welds of sample welded parts (steel and aluminum) shall have a liquid penetrant weld inspection performed in accordance with ASTM E 165, method B, procedure B-1. Each evidence of weld crack or grass porosity on any weld shall constitute a defect. Any crack in the base amaterial shall be cause rfor rejection of the lot. The lot size shall be all of each like item offered for inspection at one time. The sample size shall be one welded item of each type. The inspection level shall be S-3 and the acceptable quality level (AQL) , expressed in terms of defects per hundred units, shall be 15.

4.4.3 End item visual examination. The end items shall be examinedfor the defects listed in table I. The lot size shall be expressed in uni of wecge kits. The sample unit shall be one complete wedge kit. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Workmanship	Extraneous material such as dirt, grease, rust		201
	Rough tool, die, or grind marks		202
	Dents or scratiches not affecting serviceability		203
	Dents or scratchess not defects serviceability	101	

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TABLEL. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Wordmanship (cont'd)	Any sharp edge, corner, burr, or metal liver	102	
	Any extra hole	103	
Plating	Incomplete, not as specified	104	
Pinted surfaces	Film incomplete, nonadgerent, peeling	105	
	Film not uniform, not smooth		204
	Film contains creeps, wrinkles, sags, runs, imbedded dirt		205
	Not color specified		206
Welding	Missing, imcomplete, irregular bead, non-uniform cross section	106	
	Any crack, burn hole, under-cutting overlapping	106	
	Scale, splatter not removed		208
Threaded fasteners	Not type specified	108	
	Broken, cracked, stripped	109	
	Nuts, bolts, turnbuckles not tight		208
Construction	Any component missing or not properly located	110	
	Any component fractured or malformed	111	
Marking	Missing, incorrect, illegible, misplaced, or otherwixe not as specified		209

4.4.4 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified on the drawings. Only those dimensions that can be evaluated without damaging or disassembling the end items shall be examined. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of wedge kits. The sample kit shall be one wedge kit. The impaction level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

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4.4.5 Packaging examinatin. The fully packaged end items shall be examined for the degects listed below. The lot size shall be expressed in units of shipping containers. teh sample unit shall b4e one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Markin (exterior and interios)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequte applicaiton of components, such as: imcomplete sealing or closure of flap. improper taping, loose strapping, or inadequate stapling Bulged or distroted container
Content	Number per container is more or less than required

5. PACKAGING

5.1 Preservation. The level of preservation shall be as required by the procuring activity in accordance with MIL-STD-2073-1.

5.2 Packing. The level of packing shall be as required by the procuring activity in accordance with MIL-STD-2073-1.

6. NOTES

(This section contains information of a general or explanatory naure that may be helpful, but is not mandatory.)

6.1 Intended use. The edge kit, shen assumed with two 8 foot plateforms, makes a structure which attaches to the cargo ramp of a C-130 aircraft, providing a platform which allows kindles to exit by gravity.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specificaticn.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. Applicable levels of preservation and and 5.2).

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6.3 First article. When a first article is required it shall be inspected and approved under the appropriate provisions of Federal Acquisition Regulation FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Subject term (key word) listing.

Airdrop
Platform, air drop

6.5 Changes from previous issue. Marginal notatins are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army-GL
Air Force -99

Review activity:

Army-AV

Preparing activity:

Army-GL

(Project 1670-0803)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-W-44027A	2. DOCUMENT DATE (YYMMDD) 93-05-12

3. DOCUMENT TITLE
Wedge Kit, Ramp Bundle Delivery System, C-130 Aircraft

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME U.S. Army Natick RD&E Center	b. TELEPHONE (Include Area Code) (1) Commercial: 508-651-4531 (2) AUTOVON: 256-4531
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SATNC-UXT Natick, MA 01760-5017	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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